

Rupjyoti Barman

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Summary

.NET Full Stack Developer with **3+ years** of experience building enterprise web applications for **Hewlett-Packard** using **C#, ASP.NET Core** (MVC, Razor Views, Web API), **Entity Framework Core**, **SQL Server**, **JavaScript**, **AJAX**, and **Bootstrap**. Skilled in designing **Controller-Service-Repository** layered architecture, **Role-Based Access Control (RBAC)**, and **Factory Pattern** with dependency injection, backed by **MSTest** and **Moq** unit testing. Hands-on experience consuming native C++ hardware APIs via **C# P/Invoke** wrappers, **WMI** monitoring, and Windows APIs (**WUA**, **HPCMSL**). Collaborates with global engineering teams across India, Taiwan, and the US.

Skills

- **Languages:** C#, C++, SQL, JavaScript, Python
- **Frameworks & Libraries:** ASP.NET Core (Web API, MVC), Entity Framework Core, WPF, Bootstrap, jQuery, LINQ
- **Windows APIs & Interop:** WMI, ManagementEventWatcher, WUA API, HPCMSL, P/Invoke
- **Architecture & Design:** SOLID Principles, OOP, Factory Pattern, Dependency Injection, Controller-Service-Repository Pattern
- **Database, DevOps & Tools:** SQL Server, PostgreSQL, EF Core (Code-First, Migrations), CI/CD, Azure Pipelines, Git, NuGet, Visual Studio, Postman, Jira, Confluence, Serilog
- **Testing & Security:** MSTest, Moq, JWT Authentication, RBAC

Experience

Software Developer, BoldTek India Pvt. Ltd. - Guwahati

June 2023 - Present

Client: Hewlett-Packard (HP)

- Engineered a **Defect Code module** (full CRUD, auto-incrementing error code generation, version history tracking) and a **Release Tracking Dashboard** (dynamic multi-entry JS-cloned form with bulk **AJAX** submission) within an internal **ASP.NET Core MVC** platform using **Razor Views**, DTO/ViewModel pattern, **Controller-Service-Repository** architecture with interface-based dependency injection, **EF Core Code-First**, and **SQL Server**, replacing manual Excel-based release documentation.
- Built a **MyIssues dashboard** with **3-tier RBAC** (Viewer: read-only; ODM: leverage, test, and submit configurations for approval; Developer: create, edit with instant approval), Admin holding privilege to delete entries; implemented client-side **JS regex validation** enforcing key-value format, migrating **1000+ configuration entries** through the new validation logic; delivered UI enhancements using **Bootstrap** and **JavaScript**, with structured exception handling and **Serilog** logging.
- Led the India team in migrating from a **shared RPC-based Windows Service architecture** to a **decoupled three-layer NuGet-based architecture** (Business Logic, C# Interop Wrapper via P/Invoke, Native C++ DLL) with **Factory Pattern**-based dependency injection; achieved **100% elimination of third-party release delays** (previously 10–15 days) and **reduced diagnostic development cycle** from 4 to 3 sprints (40 to 30 working days, **25% reduction**) by removing 2 redundant architectural layers, resolving cross-team exceptions and sleep/hibernate-related service failures in **HP PC Hardware Diagnostics**.
- Developed **Business Logic** and **C# Interop Wrapper** (via P/Invoke) for **6 diagnostic modules** (**Storage:** DRV Read/Verify, SMART Health Monitor; **Audio:** Playback Test, Record Test; **Network:** NIC IRQ Validation, ROM/MAC Validation) to invoke native C++ diagnostic APIs, and independently built **2 modules from scratch** to detect and update **Drivers** (via WUA API) and **BIOS** (via HP CMSL for commercial systems), consolidating all into a **centralized diagnostic backend** with zero external service dependency.
- Replaced a service-based device discovery architecture (requiring a 2–3 minute service restart per hardware change) by **engineering a real-time WMI monitoring module** (ManagementEventWatcher and WQL on Win32_PnPEntity) into the shared “**Common**” library; reduced detection latency **from 2–3 minutes to under 4 seconds (97% reduction)**, adopted across multiple hardware diagnostic APIs.
- Maintained and troubleshoot **Azure Pipelines** that trigger **NuGet package builds**, publishing the Business Logic package into the .NET Wrapper layer to produce the final consolidated NuGet package (bundling all layers including Native C++ DLL) for testing and client consumption.
- Verified diagnostic modules using a **WPF-based API testing application** and **MSTest** unit test projects across **Business Logic** and **Wrapper** layers, leveraging **Moq** for dependency mocking to ensure reliability prior to client integration.
- Partnered with cross-functional global teams (**India, Taiwan, US**) to deliver features, debug production issues, and align on diagnostic API design across release cycles.

Education

B.Tech in Computer Science and Engineering - CGPA: 8.01/10

May 2023

North Eastern Regional Institute of Science and Technology (NERIST), Arunachal Pradesh